UNIT 1 · TOPIC 5

Plant Protection Equipment

Syllabus topic: Equipment for plant protection

What a sprayer does

1. Atomizes the spray liquid into droplets of effective size.
2. Distributes the droplets uniformly.
3. Regulates the application rate.
4. Directs the spray onto the target.

Sprayers by operation and pressure

Sprayer	Pressure	Notes
Hand/hydraulic knapsack	1–4 bar (low)	Lever-operated piston/diaphragm pump, 8–16 L tank
Hand compression	3–7 bar	Pressurized tank
Foot/pedal sprayer	up to 17–21 bar	Stationary, hose + lance
Rocker sprayer	14–28 bar	Tall trees with extension lance
Power knapsack mist blower	Air blast (low liquid pressure)	35–100 µm droplets; air velocity ~100–200 m/s at outlet
Tractor-mounted boom sprayer	2–7 bar (field crops)	Hydraulic nozzles on boom, 300–500 L tank
Air-blast/air-carrier orchard sprayer	10–40 bar liquid + air stream	Air displaces and carries droplets into the canopy
ULV/CDA (spinning disc)	—	Ultra-low volume <5 L/ha; controlled droplets 60–120 µm

Application volume	Rate
High volume	>400 L/ha (field) / >1000 L/ha (trees)
Low volume	5–400 L/ha
Ultra-low volume (ULV)	<5 L/ha

Nozzles

Nozzle	Spray pattern and use
Hollow cone	Ring-shaped deposit; a swirl (whirl) plate gives tangential velocity. Insecticide/fungicide on foliage.
Solid cone	Full circular pattern.
Flat fan	Tapered-edge elliptical pattern. Herbicide on flat surfaces; overlap 30–50%.
Others	Flood (deflector/anvil), solid-stream (gun), air-induction (AI — coarse, anti-drift).

$$\text{Discharge } q \propto \sqrt{P}$$

- Doubling the pressure raises discharge by **√2 (~41%)**; droplet size falls as pressure rises.

Droplets

- Express droplet size as **VMD** (volume median diameter, µm) and **NMD** (number median diameter).
- **Uniformity coefficient = VMD/NMD** — the closer to 1, the more uniform.
- Droplets below **100–150 µm** are drift-prone.

Target	Optimum droplet size
Flying insects	10–50 µm
Foliage insects	30–50 µm
Fungicides on foliage	30–150 µm
Herbicides (drift control)	250–500 µm

Pumps and agitation (Kepner ch. 21)

- **Positive-displacement pumps:** piston/plunger (high pressure, orchard); roller (vane) — most common on low-medium pressure field sprayers; gear; diaphragm — handles abrasives/wettable powders well.
- **Non-positive pump:** centrifugal — high volume, air-carrier sprayers.

Agitation keeps suspensions and emulsions mixed:

- **Mechanical:** paddles turn at **100–200 rpm** near the tank bottom (Kepner eq. 21.1: the minimum paddle peripheral speed rises with liquid depth). Paddle tip speeds **>500 fpm** cause foaming. Mechanical agitation is more economical for high-pressure orchard sprayers (Kepner §21.15–21.16).
- **Hydraulic (jet):** return part of the pump output through solid-stream nozzles near the tank bottom, aimed **30° above horizontal**. Hydraulic agitation is most efficient at **low pressures**.

Essential circuit elements: pressure regulator/relief valve, line strainers (**50–100 mesh**), cut-off valve.

Dusters (Kepner ch. 21)

- Dusters apply dry dust (**5–25% a.i.**).
- Components, in order of flow: hopper with agitator → feed control → fan/blower (rotary) → delivery spouts.
- Types: hand rotary (fan) duster, bellows duster, knapsack power duster, tractor-mounted.
- **Advantages:** no water needed; light. **Disadvantages:** high drift; poor deposit/retention (needs dew); less effective than sprays.

Drones (basic Unit 1 level)

- Agricultural spraying drones typically carry **10–16 L** tanks, with hollow-cone/flat-fan or centrifugal nozzles.
- They work **5–7×** faster than a knapsack sprayer.
- Application rate **10–25 L/ha** (called ULV in some sources and low-volume in others).
- Flight height **1.5–3 m** above the canopy.

- Regulated in India by **Drone Rules 2021**; Kisan Drones are promoted with SMAM subsidy; DA&FW has issued SOPs for pesticide application.

Practice MCQs

Q37. For herbicide application on flat surfaces, the recommended nozzle is:

(a) Hollow cone
(b) Flat fan
(c) Solid stream
(d) Rotary

Q38. Doubling the spraying pressure increases nozzle discharge approximately:

(a) 2 times
(b) 4 times
(c) 1.41 times
(d) No change

Q39. ULV spraying refers to application rates of:

(a) <5 L/ha
(b) <50 L/ha
(c) <150 L/ha
(d) <400 L/ha

Q40. The ratio used to express droplet spectrum uniformity is:

(a) NMD/VMD
(b) VMD/NMD
(c) VMD × NMD
(d) VMD - NMD

Q41. In mechanical agitation of sprayer tanks, paddles usually rotate at:

(a) 10–50 rpm
(b) 100–200 rpm
(c) 500–800 rpm
(d) 1000–1500 rpm

Q42. In hydraulic (jet) agitation, nozzles near the tank bottom are aimed:

(a) Vertically up
(b) About 30° above horizontal
(c) Horizontally
(d) 45° downward

Q43. Most efficient hydraulic agitation is obtained at:

(a) High pressure
(b) Low pressure
(c) Medium pressure only
(d) Independent of pressure

Q44. Which pump handles wettable-powder suspensions with least wear problems?

(a) Gear pump
(b) Diaphragm pump
(c) Roller pump
(d) Piston pump without liners

Q45. Droplet sizes most prone to spray drift are below about:

(a) 400 µm
(b) 250 µm
(c) 100–150 µm
(d) 600 µm

Answers

Q37 — (b) Flat fan

Q38 — (c) 1.41 times — *$q \propto \sqrt{P}$, so doubling P multiplies q by $\sqrt{2} \approx 1.41$.*

Q39 — (a) <5 L/ha

Q40 — (b) VMD/NMD

Q41 — (b) 100–200 rpm

Q42 — (b) About 30° above horizontal — *Kepner §21.16 — jets 1–2 in. above the tank bottom, aimed 30° above horizontal.*

Q43 — (b) Low pressure — *Kepner §21.16 — low pressures give the most efficient hydraulic agitation.*

Q44 — (b) Diaphragm pump

Q45 — (c) 100–150 µm